

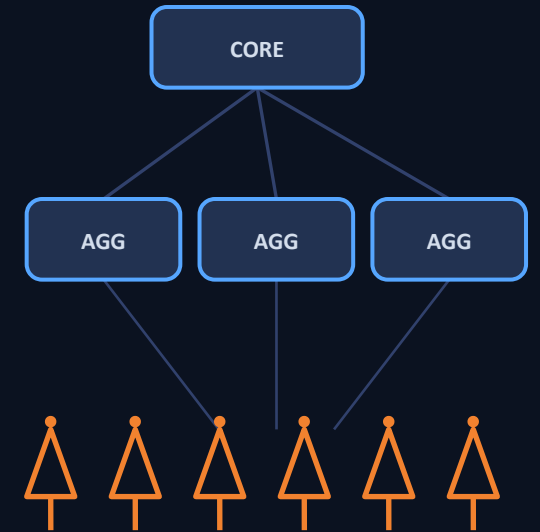
# RAN & Transport Isolation Module

Rank the sites, nodes and transport paths most impacted by transport-network problems — and tell you which are transport vs RAN, with the evidence.

IP Transport Analytics

Impact Analytics Dashboard

Phase-2 Audit Prioritisation



Synthetic-data build · standalone web app · [github.com/manjunathsept11/ran-transport-isolation](https://github.com/manjunathsept11/ran-transport-isolation)

# Correlate experience with transport — then rank and attribute

Ingest and correlate end-user experience metrics with transport-layer telemetry to identify and rank the sites, nodes and transport paths most impacted — and separate transport-attributed impairment from RAN-attributed impairment using serving-cell context and radio signal quality.

5

data feeds  
stitched

6

analytics  
modules

75–100

sites for the  
Phase-2 audit

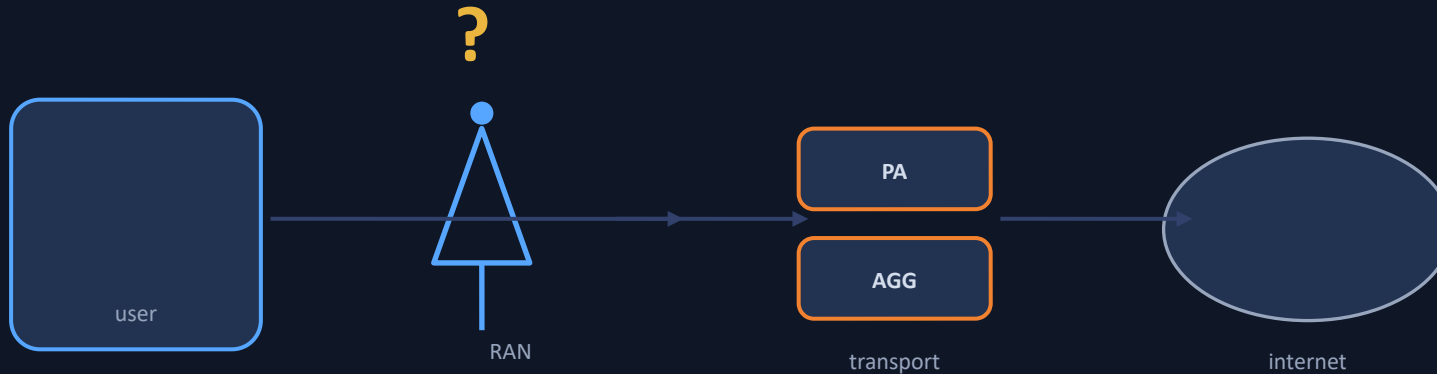
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LLM calls in  
the runtime

## Deliverables

- Impact analytics dashboard ranking sites, nodes and transport paths
- Headline KPIs: TCP RTT, TCP Fail %, Ookla throughput, VoNR MOS, YouTube QoE
- Prioritised list of the top 75–100 sites for a Phase-2 field audit (CSV / Excel)
- Per-site root-cause indicators and a recommended technician action
- Transport-attributed impairment distinguished from RAN-attributed

# Users complain. Whose problem is it?



## Looks like RAN

- Coverage hole / poor RSRP
- Interference, low RSRQ
- PRB exhaustion at busy hour
- Sleeping sector, VSWR / feeder fault

## Looks like Transport

- Fibre degradation, microwave fade
- Backhaul congestion at busy hour
- SFP / CRC errors, queue drops
- Routing flap, MTU black-hole

*Same symptom — high latency, buffering, dropped calls — two different owners, two different fixes.*

# Manual, siloed, and unmeasurable

## ● Siloed data

Ookla, N3 probes, TWAMP, SevOne each live in their own tool — no single correlated view.

## ● Manual triage

Engineers eyeball KPI graphs site by site. It does not scale to a whole market.

## ● Finger-pointing

RAN and transport teams each say 'not us'; tickets bounce between them for days.

## ● No prioritisation

Which 80 sites do we audit first? Today that answer is gut feel.

## ● No ground truth

There is no way to check whether the triage call was actually right.

## ● Slow Phase-2 audits

Shortlisting sites for field work takes weeks of cross-team analysis.

# Five feeds, one timeline



## Ookla SpeedTest

*user experience + radio*

UL/DL throughput, loaded latency, RSRP, RSRQ



## N3 probe EDRs

*session — the split point*

TCP client RTT, TCP server RTT, TCP fail %, retransmissions, VoNR MOS



## YouTube / Audio QoE

*application*

video / audio MOS, rebuffer ratio, startup delay



## TWAMP

*pure transport (active probes on each link)*

one-way / round-trip delay, frame loss, jitter



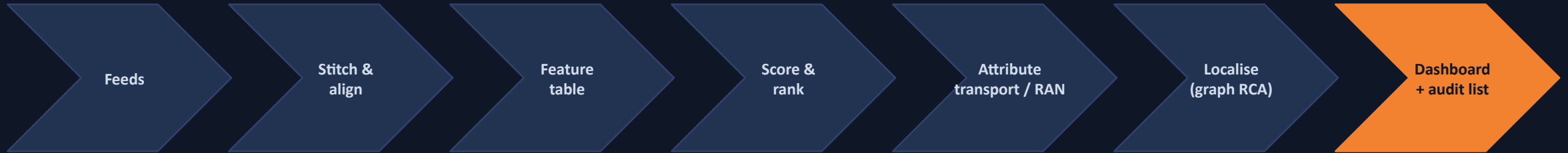
## SevOne

*transport devices*

router queue depth, discards, interface utilisation, CRC errors

*Joined on site · cell · transport-path · 5-minute bin.*

# One pipeline, batch now, streaming later



## Now — batch

- Synthetic generator writes the SQLite warehouse + Parquet raw layer
- na generate → na analytics → na report (or the Settings page)
- Dashboard reads pre-computed serving tables — always fast

## Live — streaming (Phase 2)

- The five feeds land continuously via the ingestion layer (Ramoji)
- 5-min rollups on arrival; analytics recompute on a schedule (e.g. hourly)
- Same serving tables, same dashboard — plus alerting on new priority sites

SQLite warehouse · FastAPI backend · React dashboard · background job runner

# Transport vs RAN — the tell

## 1 The RTT split

TCP client RTT = radio + the whole transport path (round trip). TCP server RTT = internet only, site-independent — the control.

Client RTT up while server RTT flat → access / transport, not the internet.

## 2 Signature match

Radio: RSRP / RSRQ degraded, PRB near 100 %, usually one sector. Transport: TWAMP loss / jitter, SevOne queue / CRC, per-link delay — the transport layer's own instruments.

Pure-transport instruments lit, radio clean → transport.

## 3 Common cause

A single radio fault cannot degrade 18 sites at once. A shared backhaul link does it every time.

'% of sibling sites on the shared uplink also degraded' is the strongest tell.

transport

RAN

shared

unclear

+ a calibrated confidence score

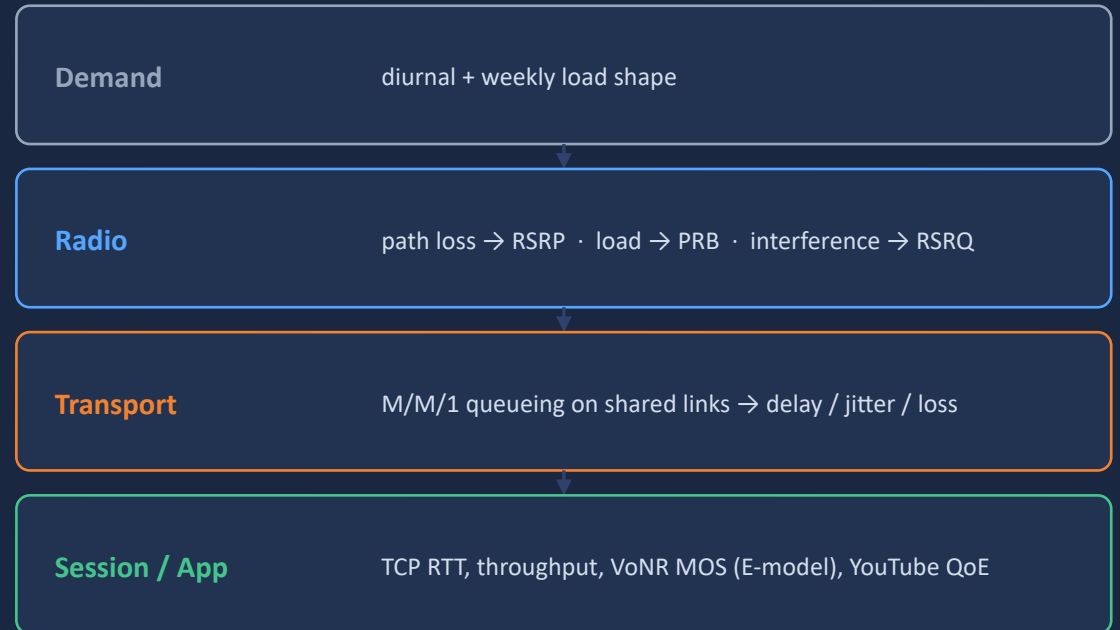
*Any one alone is weak — together they give a class and a confidence.*

# The data is synthetic — on purpose

## Why

- Data-pipeline engineering is not in scope yet — real feeds arrive in Phase 2 (Ramoji).
- A simulator lets us build and validate the analytics against known answers.
- Labelled fault injection → every record has a ground-truth cause.
- So precision / recall of the attribution is a real, reported number — not a guess.

## The layered causal model



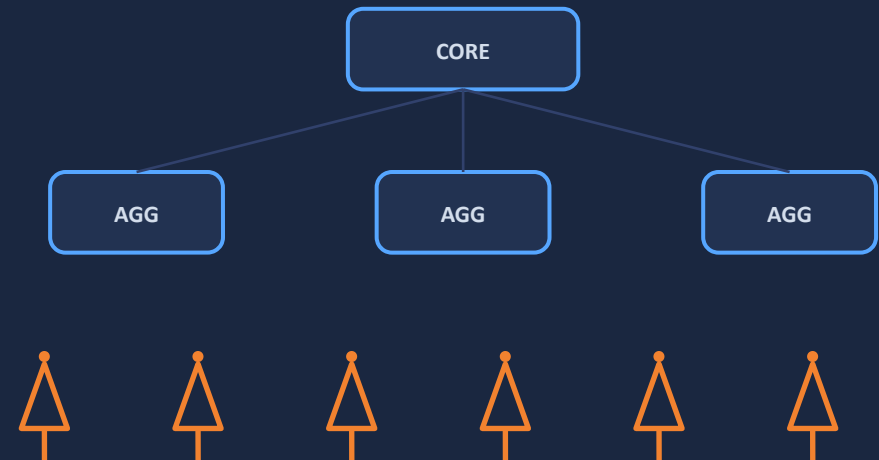
*Every metric is derived from shared latent state — so the KPI correlations are physical, not bolted-on noise.*



# Six stages — seeded and reproducible

- 1 Topology**  
market → regions → sites → cells; routers; links; per-site 3-hop path
- 2 Incidents**  
explicit + Poisson auto faults, each resolved to targets & a window  
— GROUND TRUTH
- 3 Causal metric model**  
per 5-min bin: demand → queueing → radio → session/app KPIs
- 4 Raw sessions**  
Poisson count of individual Ookla tests / QoE sessions per site-bin
- 5 Hourly rollups**  
5-min parquet → hourly; transport rolled along each site's path
- 6 Load**  
SQLite warehouse (dims, ground truth, rollups, sessions)

## 3-tier transport topology



PA (pre-agg) → AGG uplinks are shared by whole clusters

**one link fault → a whole cluster degrades → the common-cause signal**

~1,200 sites · ~1,450 links · 5-min granularity · 1–2 weeks · same seed → identical dataset

# Statistical, ML and graph — explainable first

## Baselining

robust trailing median + inter-decile range; morphology + load peer-group z-scores

## Impact scoring

weighted multi-criteria composite; session- and user-impacted weighting

## Attribution

rule engine ∪ LightGBM (gradient-boosted trees) + SHAP; isotonic-calibrated ensemble

## Anomaly

STL residual + Isolation Forest + PCA reconstruction error; time + topology clustering

## Correlation

Spearman, partial correlation (precision matrix), lead / lag cross-correlation

## Variability

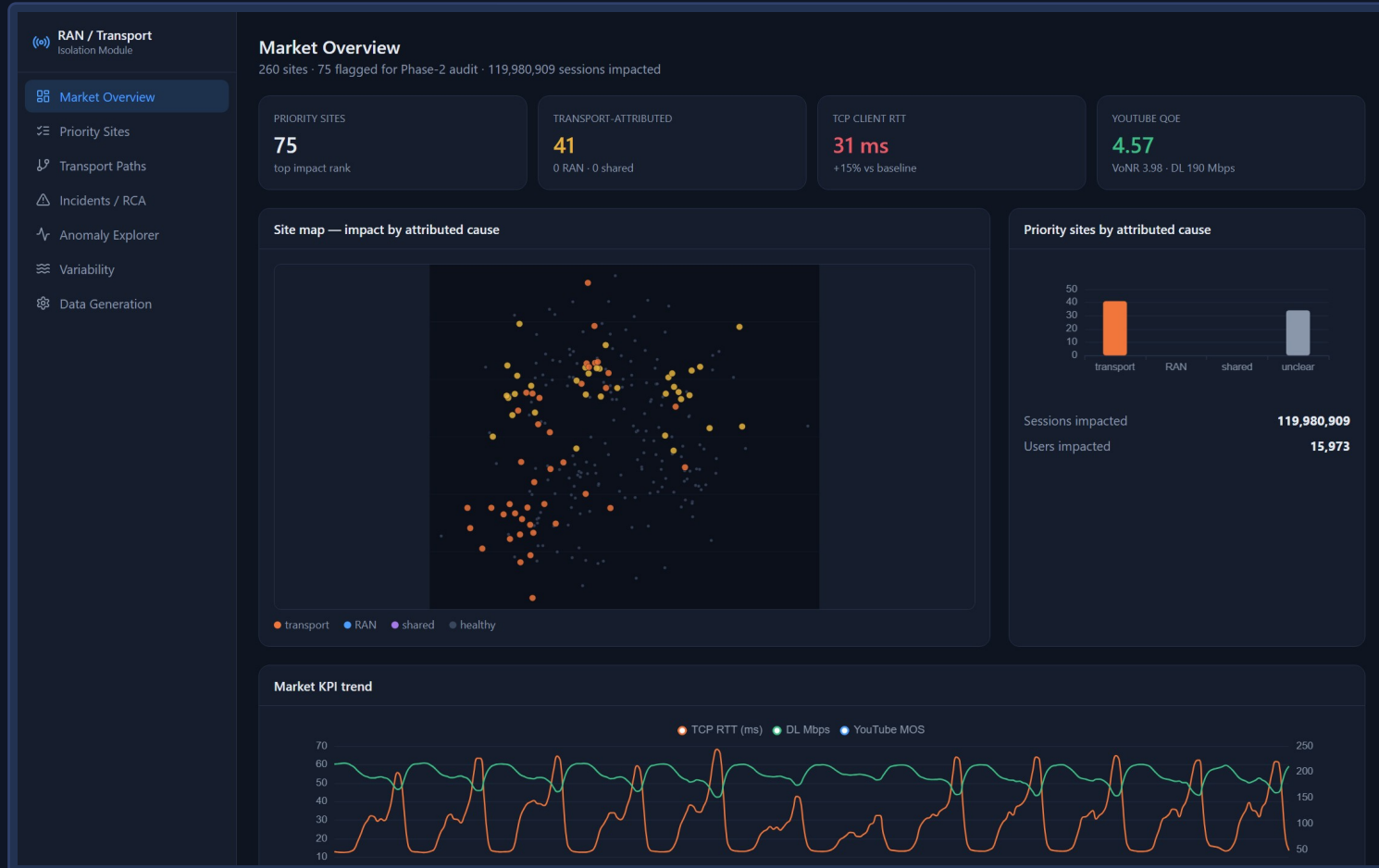
variance components, coefficient of variation, PSI drift (week over week)

## Root cause

networkx transport graph + minimal hitting set (Noisy-OR fault localisation)

**Explainable first: hand rules + SHAP + plain-English evidence. No LLM in the runtime.**

# Market overview → the priority list



Market Overview — site map coloured by attributed cause, KPI tiles vs baseline, attribution split

**Priority Sites**  
75 of 75 shown — ranked for Phase-2 field audit

all causes all regions Audit list (xlsx)

| #  | SITE   | REGION | MORPH       | IMPACT | CAUSE     | CONF. | HEADLINE SEVERITY | SESSIONS   | WORST WINDOW |
|----|--------|--------|-------------|--------|-----------|-------|-------------------|------------|--------------|
| 1  | S00178 | R3     | dense_urban | 43.1   | transport | 0.98  | RTT Fail Thp      | 6,575,571  | 08-17 11:00  |
| 2  | S00092 | R4     | dense_urban | 38.8   | transport | 0.78  | RTT Fail Thp      | 4,318,045  | 08-14 14:00  |
| 3  | S00206 | R2     | urban       | 36.0   | transport | 0.78  | RTT Fail Thp      | 3,824,836  | 08-17 13:00  |
| 4  | S00223 | R4     | urban       | 30.5   | transport | 0.88  | RTT Fail Thp      | 4,571,492  | 08-24 12:00  |
| 5  | S00004 | R5     | urban       | 26.3   | transport | 0.78  | RTT Fail Thp      | 4,308,207  | 08-24 15:00  |
| 6  | S00002 | R4     | dense_urban | 25.9   | transport | 0.78  | RTT Fail Thp      | 10,684,444 | 08-24 15:00  |
| 7  | S00019 | R1     | dense_urban | 21.1   | transport | 0.98  | RTT Fail Thp      | 2,890,851  | 08-18 16:00  |
| 8  | S00054 | R5     | urban       | 16.9   | none      | 0.78  | RTT Fail Thp      | 3,348,759  | 08-24 16:00  |
| 9  | S00154 | R4     | dense_urban | 15.4   | transport | 0.78  | RTT Fail Thp      | 2,563,432  | 08-24 16:00  |
| 10 | S00073 | R6     | suburban    | 15.3   | none      | 0.78  | RTT Fail Thp      | 4,141,513  | 08-24 16:00  |
| 11 | S00031 | R6     | suburban    | 13.9   | transport | 0.78  | RTT Fail Thp      | 2,692,609  | 08-24 15:00  |
| 12 | S00187 | R4     | dense_urban | 13.8   | transport | 0.78  | RTT Fail Thp      | 3,472,463  | 08-24 16:00  |
| 13 | S00040 | R4     | dense_urban | 13.5   | none      | 0.79  | RTT Fail Thp      | 1,079,453  | 08-24 16:00  |
| 14 | S00177 | R1     | suburban    | 13.5   | transport | 0.98  | RTT Fail Thp      | 558,965    | 08-27 12:00  |
| 15 | S00257 | R4     | dense_urban | 13.1   | none      | 0.78  | RTT Fail Thp      | 803,067    | 08-24 16:00  |
| 16 | S00105 | R6     | urban       | 13.0   | none      | 0.79  | RTT Fail Thp      | 919,369    | 08-26 15:00  |
| 17 | S00195 | R4     | dense_urban | 13.0   | transport | 0.78  | RTT Fail Thp      | 1,724,156  | 08-24 16:00  |
| 18 | S00001 | R4     | dense_urban | 12.5   | transport | 0.88  | RTT Fail Thp      | 4,233,814  | 08-24 14:00  |

Ranked top 75–100 · filter by cause / region

One-click export to the Phase-2 audit list (.xlsx)

Priority Sites — the ranked audit shortlist

# Per site: why this attribution



**Rule evidence**  
plain-English, threshold-triggered — no model

**SHAP**  
which model features drove the call

**Radio vs Transport panels**  
side by side, with anomaly markers

**Ground truth**  
the injected fault, shown in demo mode

# Root cause — localised to the link



# Who uses it — and how

## ● NOC / service assurance

daily triage — which sites to escalate, to whom

## ● IP transport operations

confirm common cause; target the link, not 18 sites

## ● RAN optimisation

filter out transport noise; focus on real radio issues

## ● Network planning

recurring congestion → capacity upgrades

## ● Field audit team

the prioritised list + the recommended checks

### The investigation — six clicks

- 1 Overview — is there a cluster?
- 2 Priority Sites — filter to transport
- 3 Site Detail — evidence + confidence
- 4 Transport Paths — siblings on the link?
- 5 Incidents / RCA — already localised?
- 6 Export the Phase-2 audit list

# Standalone, controllable, explainable

## Deploy

- One command: `docker compose up` — or native (`uv` + `Node`).
- FastAPI serves the API and the dashboard on `:8000`.
- SQLite warehouse; no external services.
- Windows / Linux; VS Code tasks included.

## Control

- The Data Generation page: pick a preset, edit distributions, schedule labelled incidents.
- Hit Generate — analytics runs automatically, with live progress.
- Save your own presets; re-run analytics or rebuild the report on demand.

## Troubleshoot

- 'none' = degraded, cause unclear — don't dispatch; low confidence → corroborate on Transport Paths.
- No data → run generate + analytics.
- Can't reach it from a laptop → bind `0.0.0.0` + open the port.
- OOM → fewer sites / days.

[github.com/manjunathsept11/ran-transport-isolation](https://github.com/manjunathsept11/ran-transport-isolation)

Docs in the repo: [README](#) · [DASHBOARD\\_GUIDE](#) · [METHODS](#) · [DATA\\_GENERATION](#) · [INSTALL\\_VM](#)